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Channel circuit with zero force equalizer

Abstract

Example channel circuits, data storage devices, and methods for using zero force adaptation to equalize a read data signal based on known data are described. A known user data signal may be determined during prior read operations and used with a residue term from the equalized read data signal to adapt the tap weights for an equalizer filter using a zero force adaptation algorithm. For example, the known user data signal may be determined by a soft output detector (e.g., SOVA detector) or full or partial decoding by an iterative decoder (e.g., LDPC decoder) and fed back for adapting the equalizer.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to read/write channel circuits for data storage devices. In particular, the present disclosure relates to read/write channels using a zero force equalizer for equalization of read data signals.

BACKGROUND

(2) Data storage devices such as hard disk drives comprise a magnetic disk and a head connected to a distal end of an actuator arm which is rotated about a pivot by a voice coil motor (VCM) to position the head radially over the disk. The disk comprises a plurality of radially spaced tracks for recording user data sectors and servo sectors. The servo sectors comprise head positioning information (e.g., a track address) which is read by the head and processed by a servo control system to control the actuator arm as it seeks from track to track. The magnetic disk acts as a non-volatile storage medium for storing data.

- (3) FIG. 1 shows a prior art disk format 2 as comprising a number of servo tracks 4 defined by servo sectors 6.sub.0-6.sub.N recorded around the circumference of each servo track. In some configurations, each servo sector comprises a preamble for storing a periodic pattern, which allows proper gain adjustment and timing synchronization of the read signal, and a sync mark for storing a special pattern used to symbol synchronize to a servo data field. The servo data field stores coarse head positioning information, such as a servo track address, used to position the head over a target data track during a seek operation. Each servo sector 6; further comprises groups of servo bursts (e.g., N and Q servo bursts), which are recorded with a predetermined phase relative to one another and relative to the servo track centerlines. The phase-based servo bursts provide fine head position information used for centerline tracking while accessing a data track during write/read operations.
- (4) FIG. 2 shows a prior art mechanical configuration for a hard disk drive (HDD) 200 comprising a spindle 202 that holds a plurality of platters 204. Each platter 204 has at least one magnetic surface 206, such as a magnetic surface formatted similarly to disk format 2 in FIG. 1. Magnetic surface 206 may be configured to have digitally encoded data stored thereon as magnetized domains. A head 208 may be mounted at the end of an arm 210 controlled by an actuator 212. A read element 214 may be configured to detect magnetized portions of magnetic surface 206 and generate an analog read signal. A write element 216 may be configured to generate a timed write field to write (or erase) magnetized portions of magnetic surface 206 using an analog write signal. In some configurations, head 208 may include more than one read element 214 and/or write element 216 and HDD 200 may include multiple heads 208, arms 210, and/or actuators 212. A preamplifier (preamp or preamplifier circuit) 218 controls the read and write signals to the corresponding read and write elements (e.g., read element 214 and write element 216) of each head (e.g., head 208). Preamplifier 218 may be attached to a flex circuit 220 that provides a data and power bus connection to a printed circuit board (not shown) with other drive control circuitry, such as a disk drive controller, through a flex interface connector 222.
- (5) The disk drive controller may include a read/write channel configured to receive an analog read signal from read element 214 through preamplifier 218 and flex circuit 220. The channel or channel circuit may convert the analog read signal to a digital read signal and perform iterative data detection and decoding to recover the data previously stored to disk surface 206. In some configurations, the channel may be configured to perform data detection and decoding on a plurality of data sectors corresponding to a data track. The channel may return detected data units corresponding to data bit representations in the analog read signal. These data units may be passed from the channel to a host interface. In some embodiments, the storage device controller may include a processor, memory, firmware, and other resources for receiving the data units from the channel, providing those data units to the host (with or without additional processing), and otherwise managing and coordinating the various storage device functions and subsystems.
- (6) The data channel may include various parameters for compensating for variations in read signals that impact data detection and decoding. For example, equalization of the read signal may enable the read channel to limit inter-symbol interference (ISI) memory length in the signal and better define or sharpen the ISI in the signal. An optimal equalization target and equalizer filter may maximize the signal response while reducing the noise residue power and preventing noise residue correlations. Least-Mean-Square (LMS) is a common approach to partial response equalization, but may be prone to mis-equalization. Zero force equalization has also been attempted, but with computationally difficult and/or expensive implementations.
- (7) Technology for improved equalization of read data signals may be needed. A cost-effective implementation of a zero force equalization filter may be advantageous.

SUMMARY

- (8) Various aspects for channel circuits with zero force equalization using known data signals are described, particularly read channels configured to use a known data signal to adapt the equalization filter tap weights based on a zero force adaptation algorithm. In some configurations, a

real-time adaptive gradient algorithm may implement zero force adaptation without matrix calculations using a known data signal derived from the read channel based on a lag for determining the known data. In some configurations, an equalization target, such as a 3-tap equalization target filter that uses measurements from a current equalization target to optimally whiten the composite noise at the output of the equalization circuit.

(9) One general aspect includes a channel circuit that includes an equalizer filter and corresponding adjustment logic. The equalizer filter is configured to: receive a read data signal, filter the read data signal based on a set of adaptive tap weights, and output an equalized read data signal. The adjustment logic is configured to: receive a known user data signal and a residue data signal, and determine the set of adaptive tap weights using a zero force adaptive algorithm.

(10) Implementations may include one or more of the following features. The known user data signal may include known bit values determined by the channel circuit during a prior read operation. The channel circuit may include a soft output bit detector configured to determine bit values from the equalized read data signal, where the known bit values comprise bit values based on output of the soft output bit detector. The circuit may include an iterative decoder configured to determine data values from the equalized read data signal based on error correction codes, where the known bit values comprise bit values based on output of the iterative decoder. The residue data signal may be based on filtering the equalized read data signal to subtract an equalization target. The channel circuit may include: an equalization target filter configured to determine the equalization target based on the known user data signal and a set of equalization target tap weights; residue auto-correlation logic configured to determine residue correlations based on current tap weights for the set of equalization target tap weights; and matrix computation logic configured to calculate new tap weights for the set of equalization tap weights based on the residue correlations. The channel circuit may include re-equalization logic configured to equalize the residue data signal to unity prior to residue correlations being determined by the residue auto-correlation logic. The adjustment logic may be further configured to use a least mean squared adaptive algorithm based on the read data signal to determine the set of adaptive tap weights. The adjustment logic may be further configured to include a weighting factor to determine a first contribution of the zero force adaptive algorithm and a second contribution of the least mean squared adaptive algorithm to determine the set of adaptive tap weights. A data storage device may include the channel circuit and may include a non-volatile storage medium configured to store data.

(11) Another general aspect includes a method that includes: determining, based on a read data signal, a known user data signal determined during a prior read operation; determining, based on residue values from an equalized read data signal, a residue data signal; determining, based on the known user data signal and the residue data signal, a set of adaptive tap weights for an equalization filter using a zero force adaptive algorithm; receiving, by the equalization filter, the read data signal; filtering, by the equalization filter, the read data signal based on the set of adaptive tap weights; and outputting the equalized read data signal.

(12) Implementations may include one or more of the following features. The known user data signal may include known bit values determined during the prior read operation. The method may include determining, by a soft output bit detector, bit values from the equalized read data signal, where known bit values may include bit values based on output of the soft output bit detector. The method may include determining, by an iterative decoder, data values from the equalized read data signal based on error correction codes, where the known bit values may include bit values based on output of the iterative decoder. The method may include filtering the equalized read data signal through an equalization target filter to determine the residue data signal. The method may include: determining an equalization target for the equalization target filter based on the known user data signal and a set of equalization target tap weights; determining residue correlations based on current tap weights for the set of equalization target tap weights and the residue data signal; and calculating new tap weights for the set of equalization tap weights based on the residue

correlations. The method may include equalizing the residue data signal to unity prior to residue correlations being determined. The method may include using a least mean squared adaptive algorithm based on the read data signal to determine the set of adaptive tap weights. The method may include: determining a weighting factor for the zero force adaptive algorithm and the least means squared adaptive algorithm; determining, based on the weighting factor, a first contribution of the zero force adaptive algorithm to determine the set of adaptive tap weights; and determining, based on the weighting factor, a second contribution of the least mean squared adaptive algorithm to determine the set of adaptive tap weights.

(13) Still another general aspect includes a data storage device that includes: a non-volatile storage medium; a channel circuit including a read channel and an equalization filter; means for determining, based on a read data signal, a known user data signal determined during a prior read operation; means for determining, based on residue values from an equalized read data signal, a residue data signal; means for determining, based on the known user data signal and the residue data signal, a set of adaptive tap weights for the equalization filter using a zero force adaptive algorithm; means for filtering, by the equalization filter, the read data signal based on the set of adaptive tap weights; and means for outputting the equalized read data signal for use by the read channel to determine user data.

(14) The present disclosure describes various aspects of innovative technology capable of improving manufacturing costs, performance, and/or error rates by supporting improved read signal equalization in the read channel based on zero force equalization. The various embodiments include operations and control circuitry to overcome or at least reduce issues previously encountered in data storage devices and, accordingly, are more efficient, reliable, and/or higher performance than other data storage devices. That is, the various embodiments disclosed herein include hardware and/or software with functionality to improve data storage device cost and performance, such as by using the known data signal from a prior read operation to adapt the equalization filter tap weights based on a zero force adaptation algorithm. Accordingly, the embodiments disclosed herein provide various improvements to data storage devices and computing systems incorporating such data storage devices.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The techniques introduced herein are illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings in which like reference numerals are used to refer to similar elements.

(2) FIG. 1 is a diagram of a prior art disk format comprising a plurality of servo tracks defined by servo sectors.

(3) FIG. 2 is a diagram of an example data storage device in the form of a disk drive comprising heads actuated over disk surfaces.

(4) FIG. 3 is a block diagram of a configuration of data storage device electronics for a controller and read/write channel configured with a zero force equalizer.

(5) FIG. 4A is a block diagram of a portion of an example channel circuit with a zero force equalizer.

(6) FIG. 4B is a block diagram of a portions of another example channel circuit with a combined zero force and LMS equalizer and alternative optimal equalization target.

(7) FIG. 5 is a block diagram of example filter configurations for an equalizer circuit.

(8) FIG. 6 is a block diagram of an example filter configuration for an equalization target circuit.

(9) FIG. 7 is an example method of using and adjusting a zero force equalizer in a read channel.

(10) FIG. 8 is an example method of determining an optimal equalization target with alternative

configurations.

(11) FIG. 9 is an example method of adjusting tap weights of an equalizer using a combined zero force and LMS adaptive algorithm or circuit.

DETAILED DESCRIPTION

(12) Prior zero force (ZF) equalization has involved a two step process. First, the unequalized isolated pulse response (ISO) is extracted from the waveform using a correlation technique with the data pattern. Then, the ISO response, which is considered noiseless, may be used to construct an equalizer to a desired equalization target based on a matrix equation. This technique may work but is difficult and/or expensive to implement in hardware and is not continuously adaptive.

(13) In data storage and retrieval, adaptive ZF equalization has been directed at the read side and used receiver equalization techniques in a finite impulse response (FIR) filter. However, by viewing the equalizer as operating on the transmission (write) side of the data signal, a different approach may be used. While this approach is not physically possible for data storage and retrieval (e.g., magnetic saturation recording in an HDD), it may provide a mathematical frame work for more cost-effectively and adaptively achieving zero force equalization by using decoded user data (either tentatively determined by a soft output Viterbi algorithm (SOVA) detector or fully decoded by an iterative error correction code (ECC) decoder) as the noiseless input to the equalization filter.

(14) Prior LMS adaptive techniques use tap gradients in an FIR filter based on the correlation of two noisy terms, such as the unequalized read signal and calculated noise residue. This may be represented by:

$$(15) \quad T_i = \text{.Math. -Atten} * \text{UnEQ}_{k+i} * \text{Residue}_k$$

Where $\Delta T_{\text{sub}.i}$ is the tap gradient, Atten is the attenuation value, $\text{UnEQ}_{\text{sub}.k+i}$ is the unequalized input signal, and $\text{Residue}_{\text{sub}.k}$ is the noise residue value. The unequalized input signal and the noise residue value are both noisy terms and the correlation based on signal and on noise may result in misequalization.

(16) The proposed adaptive ZF technique uses tap gradients based on the correlation of one noiseless term, a signed signal value corresponding to 1 and 0 values (e.g., encoded as a signed signal (-1, +1)) corresponding to bit determinations made on previously equalized read data signals, and one noisy term (noise residue value). This may be represented by:

$$(17) \quad T_i = \text{.Math. -Atten} * \text{Signed}_{k+i} * \text{Residue}_k$$

Where $\text{Signed}_{\text{sub}.k+i}$ is the signed signal value from previously determined bit decisions. There is only correlation based on signal and not on noise, which may result in proper ZF adaptation.

(18) FIG. 3 shows a portion of example control circuitry 300 for a data storage device, such as a hard disk drive. In the example shown, control circuitry 300 may include one or more controllers. Controller 302 may comprise a storage device controller configured to receive host storage commands, process storage operations for writing, reading, and managing data stored to non-volatile storage media in the disk drive, such as the magnetic media disks in FIGS. 1 and 2. In some embodiments, controller 302 may correspond to a separate host interface and read/write path to a subset of disk surfaces in a data storage device with multiple controllers. In some embodiments, controller 302 may be configured to manage servo and read/write operations for one or more actuators, heads, and corresponding writer and reader elements.

(19) Controller 302 may comprise at least one processor 304, at least one memory 306, at least one host interface 308, and access to at least one buffer memory 310. Controller 302 may also comprise a read/write channel 320, and a servo controller 342 including a servo processor 344 and servo logic 346. In some embodiments, one or more of host interface 308, read/write channel 320, and servo controller 342 may be embodied in separate packages, such as application specific integrated circuits (ASICs), systems on a chip (SOCs), or other specialized circuits that interface with processor 304 and memory 306 for carrying out their respective functions. Controller 302 may include physical and electrical interfaces for connecting to buffer memory 310, a power source (not shown), preamp 322, motor controller 348, other controllers, and/or other circuitry components. In

some embodiments, the components of controller **302** may be interconnected by a bus that includes one or more conductors that permit communication among the components. For example, processor **304**, memory **306**, host interface **308**, read/write channel **320**, and/or servo controller **342** may be components attached to a printed circuit board assembly (PCBA) **350** that provides one or more layers of interconnect conductors among the components.

(20) Processor **304** may include any type of conventional processor or microprocessor that interprets and executes instructions, and may include a plurality of processors operating alone or in combination. Memory **306** may include a random access memory (RAM) or another type of dynamic storage device that stores information and instructions for execution by processor **304** and/or a read only memory (ROM) or another type of static storage device that stores static information and instructions for use by processor **304** and/or any suitable storage element, such as a system portion of a hard disk media or a solid state storage element. Memory **306** may include multiple memory packages and/or memory types operating alone or in combination. Memory **306** may be configured to store controller firmware, comprising instructions that include one or more modules or sub-modules for specific data storage device operations and processor **304** may execute those instructions, including controlling communication with other components, such as host interface **308**, buffer memory **310**, read/write channel **320**, and servo controller **342**. In some configurations, one or more features of host interface **308**, buffer memory **310**, read/write channel **320**, and/or servo controller **342** may be embodied in firmware stored in memory **306** for execution by processor **304**.

(21) In some configurations, memory **306** may be used to store one or more modules for execution during the manufacture, configuration, and/or operation of the storage device. For example, settings **312** may be used to configure one or more components of read/write channel **320** to support ZF equalization, such as equalizer **330** and/or equalization target optimizer **332**. In some configurations, settings **312** may include one or more stored configuration parameters for determining how read/write channel **320** operates. For example, settings **312** may be embodied in a configuration page or similar data structure in memory **306**. In other configurations, settings **312** may be embodied in hardware through one or more switches or registers that are controlled or populated by firmware in memory **306**. For example, known data source **312.1** may include a switch in read/write channel **320** that determines whether the known data source used by equalizer **330** is selected from the initial bit output of SOVA detector **340.1** or the decoded bit output of iterative decoder **340.2**. In some configurations, weighting factor **312.2** may include a parameter value that determines the relative weight of ZF and LMS adaptation algorithms where equalizer **330** uses combined equalization. Settings **312** may be further described below with regard to the circuits or logic for implementing equalizer **330** and/or equalization target optimizer **332**.

(22) Host interface **308** may include any transceiver-like mechanism that enables the data storage device to communicate with other devices and/or systems, such as a host system for which the storage device provides data storage. Host interface **308** may comprise a host storage interface compliant with one or more storage interface standards, such as a Serial Advanced Technology Attachment (SATA) interface, a Small Computer System Interface (SCSI), serial attached SCSI (SAS), peripheral computer interface express (PCIe) (e.g., Non-Volatile Memory Express (NVMe)), etc., for connecting host interface **308** to peripheral interface or network port.

(23) Buffer memory **310** may include a RAM, flash, or another type of dynamic storage device for storing host data and other information in transit between the storage media of the storage device and the host (via host interface **308**). In some embodiments, buffer memory **310** is a separate memory device from memory **306** and the disk surfaces or other non-volatile memory of the data storage device.

(24) Read/write channel **320** may include one or more specialized circuits configured for processing binary data to be written to the disk surfaces using an analog write signal and processing the analog read signal from the disk surfaces back into binary data. For example, read/write

channel **320** may include a write path or write channel comprised of various data scramblers, run-length limited (RLL) encoders, iterative error correction code (ECC) encoders, precompensation circuits, and other data or signal processing components. Read/write channel **320** may include a read path or read channel comprised of various amplifiers, filters, equalizers, analog-to-digital converters (ADCs), soft information detectors, iterative ECC decoders, and other data or signal processing components. The write channel components may comprise a write channel circuit and the read channel components may comprise a read channel circuit, though the circuits may share some components. Read/write channel **320** may provide the analog write signal to and receive the analog read signal from preamp **322**, which controls and amplifies signals to and from the heads. Binary data for recording to the storage medium may be received by read/write channel **320** from controller firmware and decoded data from read/write channel **320** may be passed to controller firmware and/or directed to buffer memory **310** for communication to the host.

(25) In some configurations, read/write channel **320** may include an analog front end **324** configured to receive the analog read signal from preamp **322** and convert it into a digital read signal for processing by other components of read/write channel **320**. For example, analog front end **324** may include an ADC **326** that receives an analog data signal from preamp **322** and generates a digital signal for use by other components of read/write channel **320**. ADC **326** may sample the analog read signal at a predefined channel baud rate to determine digital signal values corresponding to a digital read signal waveform. In some configurations, analog front end **324** may include an ADC buffer memory configured to temporarily store the digital read signal generated by the ADC before it passed to other components, such as pre-equalizer **328** and/or equalizer **330** in the read path to iterative detector **340**.

(26) In some configurations, read/write channel **320** may include pre-equalizer **328**. Pre-equalizer **328** may include a signal processing circuit that precedes equalizer **330** in the read path of read/write channel **320** and may include a filter configured to reduce the size of equalizer **330**. For example, pre-equalizer **328** may be configured with a fixed set of tap weights that reduce a non-varying portion of the signal equalization and thereby reduce the operating range needed for equalizer **330**. Once the tap weights for pre-equalizer **328** are determined they may be adapted less frequently than equalizer **330**, while reducing the hardware and/or time necessary for equalizer **330** to adapt to operating read data signals.

(27) In some configurations, read/write channel **320** may include equalizer **330**. Equalizer **330** may include an equalization circuit and/or corresponding hardware and firmware configured to equalize the digital read data signal prior to processing by iterative detector **340**. For example, equalizer **330** may receive a digital read signal from ADC **326** directly or through one or more other signal processing components, such as pre-equalizer **328**, and output an equalized read data signal to one or more other components in the read path, such as iterative detector **340**. In some configurations, equalizer **330** may be configured as a partial response filter, such as a finite impulse response filter, controlled by a series of tap weights that determine how an incoming read data signal will be filtered to the equalized read data signal. An example filter configuration is shown in FIG. 5. Equalizer **330** may include tap adjustment logic configured to adapt the tap weights of the equalizer filter.

(28) In some configurations, equalizer **330** may include a ZF equalizer circuit **330.1** as tap adjustment logic configured to adapt the tap weights based on a zero force adaptive algorithm. For example, ZF equalizer circuit **330.1** may receive the read data signal, a known data signal (e.g., signed bit signal from prior detection/decoding of read data) for that read data signal, and a residue value for the read data signal after equalization through the current tap weights of equalizer **330** and corresponding to the difference between the equalized read data signal and an equalization target. ZF equalizer circuit **330.1** may apply a zero force adaptive algorithm to determining changes to the set of tap weights based on the inputs. For example, ZF equalizer circuit **330.1** may implement an adaptive algorithm in the general form of:

$$(29) \text{ .Math. -Atten} * \text{Signed}_{k+i} * \text{Residue}_k$$

The configuration and operation of ZF equalizer circuit **330.1** may be further described below with regard to FIG. **4A**.

(30) In some configurations, equalizer **330** may include a combined ZF/LMS equalizer circuit **330.2** as tap adjustment logic configured to adapt the tap weights based on a combination of a zero force adaptive algorithm and an LMS adaptive algorithm. For example, ZF/LMS equalizer circuit **330.2** may receive the read data signal, a known data signal (e.g., signed bit signal from prior detection/decoding of read data) for that read data signal, and a residue value for the read data signal after equalization through the current tap weights of equalizer **330** and corresponding to the difference between the equalized read data signal and an equalization target. A zero force adaptive algorithm may operate substantially as described above for ZF equalizer **330.1** to generate a ZF portion of the tap weight adjustments, while the LMS adaptive algorithm may implement an adaptive algorithm in the general form of:

$$(31) \quad T_i = \text{.Math. -Atten} * \text{UnEQ}_{k+i} * \text{Residue}_k$$

to generate an LMS portion of the tap weight adjustments. In some configurations, the relative contributions of the two adaptive algorithms may be determined by a weighting factor, such as weighting factor **312.2**. The resulting adaptive algorithm may be in the general form of:

$$(32) \quad T_i = \text{.Math. -Atten} * ((p * \text{Signed}_{k+i}) + ((1 - p) * \text{UnEQ}_{k+i})) * \text{Residue}_k$$

The weighting factor p determines the relative contributions of the ZF term based on the known bit data signal and the LMS term based on the unequalized input read data signal.

(33) In some configurations, read/write channel **320** may include an equalization target optimizer **332**. Equalization target optimizer **332** may include logic and/or a circuit for periodically determining the equalization (EQ) target used to determine the residue term or signal for adapting equalizer **330**. The optimal EQ target may be based on minimizing, as well as orthogonalizing, the noise residue after the equalizer. Read/write channel **320** may operate based on a current EQ target during read operations and equalization target optimizer **332** may be initiated periodically to determine or adjust the EQ target to a new EQ target from the current EQ target and resulting noisy residue values/signal. In some configurations, the current EQ target may be represented by a filter and corresponding set of equalization tap weights and the current EQ target (values corresponding to those tap weights) may be used as an input to EQ target optimizer **332** for determining changes to the tap weights representing a new EQ target. The residue terms from prior operation of equalizer **330** (using the current EQ target) may be used as input to EQ target optimizer **332**. In some configurations, equalization target optimizer **332** may use a combination of unity equalizer **332.1**, auto-correlation logic **332.2**, and target compute logic **332.3** to generate parameters for a new EQ target, such as a set of tap weights used for filtering the equalized data signal to determine the noisy residue signal. Equalization target optimizer **332** may operate “offline” to periodically recalibrate the optimal EQ target based on a set of recently read data and corresponding noise residue, rather than attempting real-time adjustment of the EQ target.

(34) In some configurations, equalization target optimizer **332** may re-equalize the current equalizer residue to a unity target without inter-symbol interference (ISI) using unity equalizer **332.1** or similar re-equalization logic. For example, unity equalizer **332.1** may generate unity non-ISI equalized noise residue that corresponds to the colored noise from the unity equalization target. Re-equalizing to unity may simplify further calculation of the optimal EQ target based on measures of noise statistics. For example, the unity noise residue may be used to adapt an adaptive equalization target filter, such as the 3-tap filter shown in FIG. **6**, to find optima tap coefficients that whiten the composite output noise. Re-equalization to unity may allow EQ target optimizer **332** to use an adaptive gradient method from the unity equalization to avoid matrix calculation or use residue auto-correlation logic **332.2** and a simplified matrix (compared to using the current EQ targets without re-equalization). For example, auto-correlation logic **322.2** may be simplified to use

a 2×2 matrix and simplified matrix computation logic for $R_{xx}(\tau)$ =residue auto-correlation in the general form of:

$$(35) \begin{bmatrix} R_{xx}(0) & R_{xx}(1) \\ R_{xx}(1) & R_{xx}(0) \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \begin{bmatrix} -R_{xx}(1) \\ -R_{xx}(2) \end{bmatrix}$$

$T_{sub.0}$ equals 1 and the changes in tap weights (see example filter configuration in FIG. 6) may be adapted using:

$$(36) T_i = \text{Math. } -\text{Atten} * CN_{k+i} * WN_k$$

Where $CN_{sub.k}$ is the colored noise residue from the current EQ target and unity non-ISI equalization target and $WN_{sub.k}$ is a composite noise resultant from being equalized to the new optimized equalization target. An output set of tap weights for configuring the equalization target used by equalizer **330** may be generated from the adaptive gradient method or residue auto-correlation after unity by target compute logic **332.3**.

(37) In some configurations, equalization target optimizer **332** may not re-equalize to unity, measure noise statistics, and use auto-correlation matrix **332.2** based on a larger scale matrix calculation. For example, the noise residue signal from equalized read data signals may generate direct noise statistics across a range of possible tap weight correlations. A larger-scale auto-correlation matrix may be processed using matrix computation logic to determine adjustments to the EQ target tap weights. For example, $R_{sub.xx}(\tau)$ may equal residue in auto-correlation, $R_{sub.yy}(\tau)$ may equal residue out auto-correlation, and $R_{sub.tt}(\tau)$ may equal the current EQ tap auto-correlation. The resulting matrix calculations may be based on the general form of:

$$(38) [R_{tt}(\) \text{convolutional matrix}] * [R_{yy}(\)] = [R_{xx}(\)]$$

$R_{sub.yy}(\tau)$ may use the same unity matrix calculation shown above with regard to the unity re-equalized approach. The convolutional matrix may be solved based on the residue in and residue out correlations to determine tap weight changes for an optimized EQ target relative to the current EQ target.

(39) In some configurations, read write channel **320** may include an iterative detector **340** configured to receive read data from the read heads and use iterative bit detection and ECC processing to decode the received read data into decoded data for further processing by controller firmware and/or communication to the host. For example, iterative detector **340** may include one or more bit detectors **324.1**, such as soft output Viterbi algorithm (SOVA) detectors, and one or more iterative decoders **324.2**, such as low density parity check (LDPC) decoders operating on multi-bit encoded symbols to decode each sector of data received by read/write channel **320**. Iterative detector **340** may receive a digital read signal from ADC **326** in analog front end **324**. In some configurations, iterative detector **340** may determine, with varying levels of accuracy or certainty, the recorded bits in the read data signal corresponding to user data. For example, bit detector **324.1** may output soft information predicting the most likely bit value for each bit position in the read data signal and iterative decoder **324.2** may output a completely decoded data sector after iterative processing that corresponds to the bit values for each bit position in the read data signal for the sector. As described above, either of these sets of known data may be used to generate a known data signal that represents the channel's determination (interim or final) of each bit value. For example, a known data signal may map 1s and 0s from the user data to +1 and -1 values corresponding to the noiseless signal value for the corresponding read data.

(40) Servo controller **342** may include one or more specialized circuits configured to process servo data, such as position error signals, from the disk surfaces and providing a control signal to position the actuators in a closed-loop control system. Servo controller **342** may also receive commands from processor **304** for positioning operations, such as seek, track follow, load, unload, sweep, idle, and other actuator positioning operations. Servo controller **342** may also implement servo error recovery processes for recovering from servo errors. In some embodiments, servo controller **342** may include servo processor **344** and servo logic **346** (stored in a servo memory). For example,

servo processor **344** may be a dedicated processor circuit and servo logic **346** may be firmware stored in RAM associated with the dedicated processor to provide dedicated computing resources for managing the servo functions. Servo controller **342** may receive servo signals read from the disk surface using preamp **322** and provided to servo controller **342**. Servo controller **342** may provide servo control signals to motor controller **348** and motor controller **348** may control one or more actuator VCMs and/or a spindle motor for rotating the disk stack.

(41) FIG. **4A** shows an example configuration of data storage device electronics **400** for a read/write channel configured with an equalizer circuit **410** using a ZF adaptive algorithm as described above and EQ target optimization without equalization to unity. Note that these two features need not be used together and are shown as part of the same electronics **400** for ease of description. Equalizer circuit **410** may be configured with an equalizer filter **412** that receives a read data signal **414** ($X_{\text{sub.k}}$), such as the digital read data signal from an ADC in the read path of the data storage device, and outputs an equalized read data signal **416** ($Y_{\text{sub.k}}$) for further processing by the read channel, such as processing by SOVA detector **430** and other components of an iterative data detector. k may represent a sequence of bit positions and corresponding values in one or more data signals. Read data signal **414** may be a combination **402** of user data signal **404** and a corresponding noise component **406** ($n_{\text{sub.k}}$). Equalizer circuit **410** may be configured to eliminate as much of noise component **406** as possible to assist SOVA detector **430** and other components of the read/write channel in correctly identifying the data component of read data signals **414**.

(42) In some configurations, equalizer filter **412** comprises a set of taps in a partial response filter for modifying read data signal **414**. The current set of tap weights for equalizer filter **412** determines the filtering to equalize read data signal **414**. Equalizer circuit **410** may include ZF adaptive algorithm logic **420** for dynamically adapting tap weights **418** ($T_{\text{sub.x}}$) of equalizer filter **412**. For example, each time ZF adaptive algorithm logic **420** processes a set of inputs it may output updated tap weights **418** (as a set of tap weights or tap weight changes relative to a prior set of tap weights) to adapt operation of equalizer filter **412**. ZF adaptive algorithm logic **420** may receive a residue signal **424** ($\text{Res}_{\text{sub.x}}$) and a known data signal **426** for determining changes in tap weights **418**. Known data signal **426** provides a noiseless term for zero force adaptation and residue signal **424** provides the noisy term to be reduced by adapting tap weights **418**. For example, known data signal **426** may comprise bit values $a_{\text{sub.k}+1, -1}$ determined from prior operation of the read/write channel to determine bit values from the equalized read data signal. Residue signal **424** may comprise a noisy residue signal leftover from filtering or subtracting an equalization target **422** ($\text{EqTrg}()$) (corresponding to known data signal **426**) from equalized read data signal **416**. In some configurations, ZF adaptive algorithm logic **420** may also receive read data signal **414**.

(43) In some configurations, ZF adaptive algorithm logic **420** may use known data signal **426** from a plurality of sources within the read channel. For example, known data signal **426** may be received from SOVA detector **430** as tentative user data **432**. SOVA detector **430** or a similar bit detector may process the equalized read signal through a Viterbi matrix to determine the most likely sequence of bit values in the read signal and calculate related soft information describing the likelihood of the bits in that pattern. The initial bit sequence identified by SOVA detector **430** may be generated relatively quickly after the read data signal has been equalized and, therefore, may provide more rapid feedback for adapting the tap weights of equalization filter **412**. However, those bit values are tentative and may be modified by further processing through an iterative decoder, such as an LDPC decoder, for making a final read determination of the bit sequence in equalized read data signal **416**. The full operation of channel write and read **440** may generate a more reliable set of known user data **442** and, in some configurations, ZF adaptive algorithm logic **420** may use known user data **442** as the noiseless input term for ZF adaptation of tap weights **418**. For example, channel write/read **440** operation may generate a fully decoded sector of data with bit values validated through the iterative ECC process that may then be used as known data signal **426**. In

some configuration, equalization circuit **410** may include a multiplexor or switch **428** for switching between the different sources of known data signals, such as tentative user data **432** and known user data **442**. The selection and corresponding processing delay between known data sources may be based on the rate of adaptation and/or the reliability of the adaptation and it may be desirable to switch sources for different operating periods and/or conditions (such as increasing error rates) over the life of the data storage device. In some configurations, equalization target **422** may also be based on convolving the known data values **426** as an input with the tap weights of an equalization target filter for determining residue signal **424**. Switch **428** may also be used to select between the known data sources for operating residue determinations for ZF adaptive algorithm logic **420** and/or residue determinations used for noise auto-correlation to update equalization target **422**. (44) In some configurations, electronics **400** may include logic for updating equalization target **422**. For example, residue signal **424** may be received by residue auto-correlation measurement logic **450** to measure noise terms associated with each bit position in equalized read data signal **416** for a sample of data. Residue auto-correlation measurement logic **450** may determine the noise metrics associated with the received residue signal **424**. The noise metrics may then be used for offline matrix computation of an optimal equalization target. For example, a matrix based on the current equalization target tap values may be evaluated using the resulting noise metrics to determine correlations that would improve the equalization target. Tap weight changes may be determined from the matrix computation and be used to modify the tap weights of the equalization target filter of equalization target **422**.

(45) FIG. 4B shows another example configuration of data storage device electronics **460** for a read/write channel configured with an equalizer circuit **462**. This version of equalizer circuit **462** includes ZF/LMS adaptive algorithm logic **464** that uses a combination of a ZF adaptive algorithm and an LMS adaptive algorithm. Additionally, electronics **460** includes re-equalize to unity logic **466** configured to re-equalize residue signal **424** to unity prior to processing through residue auto-correlation measurement **450** and offline matrix computation of optimal EQ target logic **452**. As mentioned regarding FIG. 4A, these features may be combined or used separately, but are shown together in electronics **460** for ease of description.

(46) In some configurations, equalizer circuit **462** may include ZF/LMS adaptive algorithm logic **464** configured for dynamically adapting tap weights **418** of equalizer filter **412**. ZF/LMS adaptive algorithm **464** may receive read data signal **414**, residue signal **424**, and a known data signal **426** for determining changes in tap weights **418**. Read data signal **414** may be input to an LMS term of the ZF/LMS adaptive algorithm. Known data signal **426** may be input to a ZF term of the ZF/LMS adaptive algorithm. Residue signal **424** may provide the residue term to be reduced through the tap weight adjustments. As described above, ZF/LMS adaptive algorithm **464** may be configured with a weighting factor for determining the relative contributions of the ZF and LMS adaptive algorithm terms to adapting tap weights **418**.

(47) In some configurations, electronics **460** may include re-equalize to unity logic **466** to pre-process residue signal **424** for determination of adjustments to equalization target **422**. As described above, re-equalizing residue signal **424** to unity (**1** or **0**) reduces the variability of noise metrics and simplifies further processing to determine tap weight adjustments for equalization target **422**. In some configurations, not shown, re-equalize to unity **466** may support an adaptive gradient method to determine changes to equalization target **422** without offline matrix computation **452**. In the configuration shown, re-equalized residue signal **424** may be used for residue auto-correlation measurement **450** and simplified offline matrix computation **452**. Regardless of method, new tap weights or changes to tap weights may be provided to an equalization target filter to update equalization target **422** in equalization circuit **462** responsive to periodic optimization of the equalization target based on a set of operating read data and prior performance of equalizer circuit **462**.

(48) FIG. 5 shows an example equalizer filter **500**, such as may be used in equalizer circuits **410**

and **462** in FIGS. 4A and 4B. Equalizer filter **500** may receive unequalized read data signal **502** (X.sub.k), process the received signal through an FIR filter configuration, and output equalized read data signal **504** (Y.sub.k). Equalizer filter **500** may include a delay line comprised of a series of delay circuits **510.1-510.n** for a filter length (L) and corresponding number of taps (n). Each delay **510** may have a corresponding tap filter **512** and associated tap weight value (T.sub.0-L-T.sub.L). As discussed above, the set of adaptive tap weights for tap filters **512** may be adapted during operation of the equalizer circuit using a ZF adaptive algorithm or a combined ZF/LMS adaptive algorithm. Summation line **514** combines the signals through each delay **510** and tap filter **512** to determine equalized read data signal **504**.

(49) FIG. 6 shows an example EQ target filter **600**, such as may be used in equalizer circuits **410** and **462**, to determine the residue signal from the equalized read data signal. EQ target filter **600** may receive the colored noise signal **602** (CN.sub.k) from the current equalization filter and referenced to a unity non-ISI equalization target and corresponding residual noise, process the received signal through an FIR filter configuration (e.g., a 3-tap FIR filter), and output a whitened noise signal **604** (WN.sub.k) or whitened composite noise resultant from being equalized to the new equalization target. EQ target filter **600** may include a delay line comprised of a series of delay circuits **610.1-610.2** for a filter length and corresponding number of taps (3). EQ target filter **600** may include a tap 0 with a tap weight of 1 that passes the input signal to summation line **614**. Each delay **610** may have a corresponding tap filter **612** and associated tap weight value (T.sub.1-T.sub.2). As discussed above, the set of adaptive tap weights for tap filters **612** may be modified during operation of the equalizer circuit using offline calculation of updated tap weights for improving the optimal equalization target. The tap weights may be convolved with the known data values in the signal (corresponding to bit values 0 or 1) to determine the equalization target values to be filtered from the equalized read data signal. Summation line **614** combines the signals through each delay **610** and tap filter **612** to determine residue signal **604**.

(50) As shown in FIG. 7, control circuitry **300** and/or electronics **400** or **460** may be operated according to an example method **700** of using and adjusting a zero force equalizer in a read channel. In some configurations, blocks **702** may be executed during read operations by a data storage device to equalize the read data signal and blocks **704** may be selectively executed to adapt the equalizer circuit using zero force equalization based on prior or in-process read operations that produce known data with which to evaluate operation of the equalizer. In some configurations, blocks **706** and/or blocks **704** may be executed by an equalizer circuit.

(51) At block **710**, a read data signal may be received. For example, the equalizer circuit may receive the digital read data signal from the ADC in the analog front end of a read/write channel.

(52) At block **712**, tap weights may be determined. For example, the equalizer circuit may include an equalizer filter configured with a set of tap weights determined from previously processed read data and adapted according to blocks **704**.

(53) At block **714**, the read data signal may be filtered to equalize the read data signal. For example, the read data signal may pass through the equalizer filter to equalize the read data signal based on the set of tap weights determined at block **712**.

(54) At block **716**, an equalized read data signal may be output. For example, the equalizer circuit may output the equalized read data signal for use in determining user data from the read data signal.

(55) At block **718**, user data may be determined from the read data signal. For example, the read/write channel may include an iterative detector for determining the bit data values in the read data signal. In some configurations, user data may be determined according to blocks **720** and **722**.

(56) At block **720**, tentative bit values may be determined. For example, a bit detector, such as a SOVA detector, may determine most likely bit values for a sequence of bit positions in the read data signal and return soft information related to the possible bit values in the read data signal.

(57) At block **722**, user data values may be iteratively determined. For example, an iterative

decoder, such as an LDPC decoder, may determine the user data encoded in the read data signal based on the soft information output from the bit detector and using iterative ECC processing for sequences of read data, such as codewords or sectors.

(58) At block **724**, user data may be output. For example, the read channel may return the user data for a data sector to data storage device firmware and/or a host requesting the data.

(59) At block **730**, a known user data signal may be determined. For example, the equalizer circuit may include tap adjustment logic that receives a known user data signal based on prior or ongoing read processing through the read channel.

(60) At block **732**, a source of known user data may be determined. For example, the tap adjustment logic may include a switch or multiplexor for determining whether tentative bit values determined at block **720** or final bit values determined at block **722** are used for the known user data signal.

(61) At block **734**, a residue data signal may be determined. For example, the tap adjustment logic may receive a residue data signal based on the noisy residue remaining after processing the read data signal through the equalizer circuit in blocks **706**. In some configurations, the residue data signal may be determined blocks **736** and **738**.

(62) At block **736**, an equalization target may be determined. For example, the equalization target may include a filter that convolves the known bit value with tap weights representing an ideal read data signal.

(63) At block **738**, the equalized read data signal may be filtered using the equalization target. For example, the equalized read data signal may pass through the equalization target filter to generate a data signal corresponding to the noisy remainder of the equalized read data signal after the ideal read data signal is removed.

(64) At block **740**, zero force correlation may be calculated. For example, the tap adjustment logic may use a zero force adaptive algorithm to process the known user data signal and the residue data signal to determine new tap weights or tap weight changes.

(65) At block **742**, the tap weights for the equalizer filter may be adjusted. For example, the tap adjustment logic may feed the tap weight changes back to the equalization filter to adjust the tap weights for the next read data to be processed.

(66) As shown in FIG. **8**, control circuitry **300** and/or electronics **400** or **460** may be operated according to an example method **800** of determining an optimal equalization target with alternative configurations. In some configurations, blocks **820-822** may use equalization to unity and an adaptive gradient algorithm to avoid matrix calculations in determining an updated equalization target. In some configurations, blocks **824-826** may use a correlation matrix from the residue data signal to determine updated tap weights for the equalization target.

(67) At block **810**, a residue data signal may be received. For example, equalization target optimizer logic may receive residue data calculated from prior read operations and an existing set of EQ target tap weights.

(68) At block **812**, residue correlation may be determined. For example, the equalization target optimizer logic may calculate noise metrics from the residue data based on correlations with the current EQ target tap weights.

(69) At block **814**, tap weights may be calculated for the equalization target. For example, based on the correlations in the noise metrics, the equalization target optimizer logic may determine changes in the tap weights to improve performance of the equalizer based on a more optimal EQ target.

(70) At block **816**, the tap weights for the equalization target may be set. For example, the equalization target optimizer logic may send the updated tap values or tap value changes to modify the tap weights of the equalization target filter.

(71) At block **820**, the residue data signal may be equalized to unity. For example, the equalization target optimizer logic may re-equalize the residue data signal to unity prior to determining noise metrics and correlations.

(72) At block **822**, an adaptive gradient algorithm may be applied. For example, the equalization target optimizer logic may use the re-equalized residue data and existing tap weights to determine changes to the tap weights that reduce the noisy residue.

(73) At block **824**, a correlation matrix may be determined. For example, the equalization target optimizer logic may determine a correlation matrix for the existing tap weights and resulting residue data.

(74) At block **826**, the matrix may be solved for updated tap weights. For example, the equalization target optimizer logic may use sets of auto-correlations to solve for tap weights that reduce the noisy residue.

(75) As shown in FIG. 9, control circuitry **300** and/or electronics **460** may be operated according to an example method **900** of adjusting tap weights of an equalizer using a combined zero force and LMS adaptive algorithm or circuit. Blocks **912-916** may determine an LMS portion of the adaptive algorithm and blocks **918-922** may determine a zero force portion of the adaptive algorithm.

(76) At block **908**, a residue data signal may be determined. For example, the tap adjustment logic may receive residue data from subtracting an equalization target for the known data signal from the equalized read data signal to determine the noise component that remains after equalization.

(77) At block **910**, a weighting factor may be determined. For example, the tap adjustment logic may be configured with or access a weighting factor value that determines the relative contribution of the zero force and LMS terms of the adaptive algorithm.

(78) At block **912**, an unequalized read data signal may be received. For example, the tap adjustment logic may use the read data signal prior to processing through the equalizer.

(79) At block **914**, an LMS correlation may be determined. For example, the tap adjustment logic may use correlations between the read data signal and the residue data signal to calculate one or more tap adjustments.

(80) At block **916**, contribution of the LMS portion of the adaptive algorithm may be determined. For example, the tap adjustment logic may use the weighting factor determined at block **910** to scale the tap adjustments determined by the LMS correlations at block **914**.

(81) At block **918**, a known user data signal may be determined. For example, the tap adjustment logic may receive known bit values determined by the read channel processing equalized data signals.

(82) At block **920**, a zero force correlation may be calculated. For example, the tap adjustment logic may use correlations between the known data signal and the residue data signal to calculate one or more tap adjustments.

(83) At block **922**, contribution of the zero force portion of the adaptive algorithm may be determined. For example, the tap adjustment logic may use the weighting factor determined at block **910** to scale the tap adjustments determined by the zero force correlations at block **920**.

(84) At block **924**, an attenuation value may be determined. For example, the tap adjustment logic may be configured with an attenuation value to control the rate at which the tap values may be adjusted.

(85) At block **926**, tap weight changes may be determined. For example, the tap adjustment logic may determine the actual tap adjustments values to be applied to the equalization filter taps based on the weighted LMS and zero force terms and the attenuation and residue values.

(86) Technology for using zero force adaptation based on known data signals for hardware-efficient adaptive equalization in a channel circuit is described above. In the above description, for purposes of explanation, numerous specific details were set forth. It will be apparent, however, that the disclosed technologies can be practiced without any given subset of these specific details. In other instances, structures and devices are shown in block diagram form. For example, the disclosed technologies are described in some implementations above with reference to particular hardware.

(87) Reference in the specification to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment is

included in at least one embodiment or implementation of the disclosed technologies. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment or implementation.

(88) Some portions of the detailed descriptions above may be presented in terms of processes and symbolic representations of operations on data bits within a computer memory. A process can generally be considered a self-consistent sequence of operations leading to a result. The operations may involve physical manipulations of physical quantities. These quantities take the form of electrical or magnetic signals capable of being stored, transferred, combined, compared, and otherwise manipulated. These signals may be referred to as being in the form of bits, values, elements, symbols, characters, terms, numbers, or the like.

(89) These and similar terms can be associated with the appropriate physical quantities and can be considered labels applied to these quantities. Unless specifically stated otherwise as apparent from the prior discussion, it is appreciated that throughout the description, discussions utilizing terms for example “processing” or “computing” or “calculating” or “determining” or “displaying” or the like, may refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.

(90) The disclosed technologies may also relate to an apparatus for performing the operations herein. This apparatus may be specially constructed for the required purposes, or it may include a general-purpose computer selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a computer readable storage medium, for example, but is not limited to, any type of disk including floppy disks, optical disks, CD-ROMs, and magnetic disks, read-only memories (ROMs), random access memories (RAMs), erasable programmable read-only memories (EPROMs), electrically erasable programmable read-only memories (EEPROMs), magnetic or optical cards, flash memories including universal serial bus (USB) keys with non-volatile memory or any type of media suitable for storing electronic instructions, each coupled to a computer system bus.

(91) The disclosed technologies can take the form of an entire hardware implementation, an entire software implementation or an implementation containing both hardware and software elements. In some implementations, the technology is implemented in software, which includes but is not limited to firmware, resident software, microcode, etc.

(92) Furthermore, the disclosed technologies can take the form of a computer program product accessible from a non-transitory computer-usable or computer-readable medium providing program code for use by or in connection with a computer or any instruction execution system. For the purposes of this description, a computer-usable or computer-readable medium can be any apparatus that can contain, store, communicate, propagate, or transport the program for use by or in connection with the instruction execution system, apparatus, or device.

(93) A computing system or data processing system suitable for storing and/or executing program code will include at least one processor (e.g., a hardware processor) coupled directly or indirectly to memory elements through a system bus. The memory elements can include local memory employed during actual execution of the program code, bulk storage, and cache memories which provide temporary storage of at least some program code in order to reduce the number of times code must be retrieved from bulk storage during execution. In some configurations, multiple processors and/or memory elements may be used alone or in combination to execute and/or store the program code.

(94) Input/output or I/O devices (including but not limited to keyboards, displays, pointing devices, etc.) can be coupled to the system either directly or through intervening I/O controllers.

(95) Network adapters may also be coupled to the system to enable the data processing system to

become coupled to other data processing systems or remote printers or storage devices through intervening private or public networks. Modems, cable modems, and Ethernet cards are just a few of the currently available types of network adapters.

(96) The terms storage media, storage device, and data blocks are used interchangeably throughout the present disclosure to refer to the physical media upon which the data is stored.

(97) Finally, the processes and displays presented herein may not be inherently related to any particular computer or other apparatus. Various general-purpose systems may be used with programs in accordance with the teachings herein, or it may prove convenient to construct more specialized apparatus to perform the required method operations. The required structure for a variety of these systems will appear from the description above. In addition, the disclosed technologies were not described with reference to any particular programming language. It will be appreciated that a variety of programming languages may be used to implement the teachings of the technologies as described herein.

(98) The foregoing description of the implementations of the present techniques and technologies has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the present techniques and technologies to the precise form disclosed. Many modifications and variations are possible in light of the above teaching. It is intended that the scope of the present techniques and technologies be limited not by this detailed description. The present techniques and technologies may be implemented in other specific forms without departing from the spirit or essential characteristics thereof. Likewise, the particular naming and division of the modules, routines, features, attributes, methodologies and other aspects are not mandatory or significant, and the mechanisms that implement the present techniques and technologies or its features may have different names, divisions and/or formats. Furthermore, the modules, routines, features, attributes, methodologies and other aspects of the present technology can be implemented as software, hardware, firmware or any combination of the three. Also, wherever a component, an example of which is a module, is implemented as software, the component can be implemented as a standalone program, as part of a larger program, as a plurality of separate programs, as a statically or dynamically linked library, as a kernel loadable module, as a device driver, and/or in every and any other way known now or in the future in computer programming. Additionally, the present techniques and technologies are in no way limited to implementation in any specific programming language, or for any specific operating system or environment. Accordingly, the disclosure of the present techniques and technologies is intended to be illustrative, but not limiting.

Claims

1. A channel circuit, comprising: an equalizer filter configured to: receive a read data signal; filter the read data signal based on a set of adaptive tap weights; and output an equalized read data signal; and adjustment logic configured to: receive a known user data signal and a residue data signal, wherein the known user data signal comprises known bit values determined by the channel circuit during a prior read operation; and determine the set of adaptive tap weights using a zero force adaptive algorithm.
2. The channel circuit of claim 1, further comprising: a pre-equalizer circuit configured to filter the read data signal based on a fixed set of tap weights prior to the equalizer filter receiving the read data signal.
3. The channel circuit of claim 1, further comprising: a soft output bit detector configured to determine bit values from the equalized read data signal, wherein the known bit values comprise bit values based on output of the soft output bit detector.
4. The channel circuit of claim 1, further comprising: an iterative decoder configured to determine data values from the equalized read data signal based on error correction codes, wherein the known bit values comprise bit values based on output of the iterative decoder.

5. The channel circuit of claim 1, wherein the residue data signal is based on filtering the equalized read data signal to subtract an equalization target.
6. The channel circuit of claim 5, further comprising: an equalization target filter configured to determine the equalization target based on the known user data signal and a set of equalization target tap weights; residue auto-correlation logic configured to determine residue correlations based on current tap weights for the set of equalization target tap weights; and matrix computation logic configured to calculate new tap weights for the set of equalization tap weights based on the residue correlations.
7. The channel circuit of claim 6, further comprising: re-equalization logic configured to equalize the residue data signal to unity prior to residue correlations being determined by the residue auto-correlation logic.
8. The channel circuit of claim 1, wherein the adjustment logic is further configured to use a least mean squared adaptive algorithm based on the read data signal to determine the set of adaptive tap weights.
9. The channel circuit of claim 8, wherein the adjustment logic is further configured to include a weighting factor to determine a first contribution of the zero force adaptive algorithm and a second contribution of the least mean squared adaptive algorithm to determine the set of adaptive tap weights.
10. A data storage device comprising the channel circuit of claim 1 and further comprising a non-volatile storage medium configured to store data.
11. A method comprising: determining, based on a read data signal, a known user data signal determined during a prior read operation; determining, based on residue values from filtering an equalized read data signal through an equalization target filter, a residue data signal; determining, based on the known user data signal and the residue data signal, a set of adaptive tap weights for an equalization filter using a zero force adaptive algorithm; receiving, by the equalization filter, the read data signal; filtering, by the equalization filter, the read data signal based on the set of adaptive tap weights; and outputting the equalized read data signal.
12. The method of claim 11, wherein the known user data signal comprises known bit values determined during the prior read operation.
13. The method of claim 12, further comprising: determining, by a soft output bit detector, bit values from the equalized read data signal, wherein the known bit values comprise bit values based on output of the soft output bit detector.
14. The method of claim 12, further comprising: determining, by an iterative decoder, data values from the equalized read data signal based on error correction codes, wherein the known bit values comprise bit values based on output of the iterative decoder.
15. The method of claim 11, further comprising: filtering, prior to the equalization filter receiving the read data signal, the read data signal through a pre-equalizer circuit comprising a fixed set of tap weights.
16. The method of claim 11, further comprising: determining an equalization target for the equalization target filter based on the known user data signal and a set of equalization target tap weights; determining residue correlations based on current tap weights for the set of equalization target tap weights and the residue data signal; and calculating new tap weights for the set of equalization tap weights based on the residue correlations.
17. The method of claim 16, further comprising: equalizing the residue data signal to unity prior to residue correlations being determined.
18. The method of claim 11, further comprising: using a least mean squared adaptive algorithm based on the read data signal to determine the set of adaptive tap weights.
19. The method of claim 18, further comprising: determining a weighting factor for the zero force adaptive algorithm and the least means squared adaptive algorithm; determining, based on the weighting factor, a first contribution of the zero force adaptive algorithm to determine the set of

adaptive tap weights; and determining, based on the weighting factor, a second contribution of the least mean squared adaptive algorithm to determine the set of adaptive tap weights.

20. A data storage device comprising: a non-volatile storage medium; a channel circuit including a read channel and an equalization filter; means for determining, based on a read data signal, a known user data signal determined during a prior read operation; means for determining, based on residue values from an equalized read data signal, a residue data signal; means for determining a set of adaptive tap weights for the equalization filter using a weighted combination of: a zero force adaptive algorithm based on the known user data signal and the residue data signal; and a least mean squared adaptive algorithm based on the read data signal; means for filtering, by the equalization filter, the read data signal based on the set of adaptive tap weights; and means for outputting the equalized read data signal for use by the read channel to determine user data.

21. A data storage device, comprising: a non-volatile storage medium configured to store data; and a channel circuit comprising: an equalizer filter configured to: receive a read data signal generated from the data stored on the non-volatile storage medium; filter the read data signal based on a set of adaptive tap weights; and output an equalized read data signal; and adjustment logic configured to: receive a known user data signal and a residue data signal, wherein the residue data signal is based on filtering the equalized read data signal to subtract an equalization target; and determine the set of adaptive tap weights using a zero force adaptive algorithm.
